SESSION 6: CORRELATION AND REGRESSION ANALYSIS

Introduction to Statistical Data Analysis Using R & RStudio

LUMUMBA W.V

Center for Data Analytics and Modeling
Chuka University

March 2026

Resource: `https://cdam.chuka.ac.ke`

Learning Objectives

By the end of this session, participants will be able to:

1. Understand the historical context and theoretical foundations of correlation and regression
2. Differentiate between correlation and causation
3. Compute and interpret Pearson, Spearman, and Kendall correlation coefficients in R
4. Formulate Simple and Multiple Linear Regression models
5. Verify regression assumptions (LINE) using diagnostic plots and tests
6. Detect and handle multicollinearity and influential observations
7. Report statistical findings in APA format

Session Outline

1. Historical Background
2. Correlation Analysis Theory
3. Correlation in R
4. Visualization of Relationships
5. Simple Linear Regression (SLR)
6. Least Squares Method
7. Regression Assumptions (LINE)
8. SLR Implementation in R
9. Multiple Linear Regression (MLR)
10. Multicollinearity & VIF
11. Model Diagnostics
12. Remedial Measures
13. Reporting & Best Practices

History: The Origins of Correlation

Francis Galton (1822–1911):

- Cousin of Charles Darwin; pioneered human intelligence measurement.
- Observed "Regression to the Mean" in heredity (tall parents $\rightarrow$ shorter children).
- Coined the term *regression* in 1885.

Karl Pearson (1857–1936):

- Developed the *Product-Moment Correlation Coefficient* (r).
- Formalized the mathematical foundation of correlation (1896).
- Established biometrics as a field.

Historical Note: Originally used for biological heredity, now fundamental to all data science.

History: Evolution of Regression

Early Applications:

- **1800s:** Astronomers used least squares (Legendre, 1805; Gauss, 1809) to predict planetary orbits.
- **1900s:** Yule used regression for social sciences (poverty vs. relief rates).
- **1920s:** Fisher formalized inference (p-values, ANOVA table for regression).

Modern Era:

- Foundation for Machine Learning (Linear Regression, GLMs).
- Extended to Logistic, Poisson, and Mixed Models.

Concept: What is Correlation?

Definition: A statistical measure that describes the extent to which two variables change together.

Key Characteristics:

- **Direction:** Positive (both increase) or Negative (one increases, other decreases).
- **Strength:** Weak, Moderate, Strong (magnitude of coefficient).
- **Form:** Linear vs. Non-linear.

Crucial Distinction: Correlation $\neq$ Causation.

- Example: Ice cream sales and drowning incidents are correlated (both driven by summer heat).

Types of Correlation Coefficients

Method	Data Type	Assumptions
Pearson (r)	Continuous, Interval	Normality, Linearity, Homoscedasticity
Spearman (ρ)	Ordinal or Continuous	Monotonic relationship, Robust to outliers
Kendall (τ)	Ordinal or Small Samples	Monotonic, Better for small n

Selection Rule:

- Normal + Linear $\rightarrow$ Pearson
- Non-Normal / Ordinal $\rightarrow$ Spearman or Kendall

Pearson Correlation Coefficient

Formula:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Properties:

- Range: $-1 \leq r \leq 1$
- $r = 1$: Perfect positive linear relationship
- $r = -1$: Perfect negative linear relationship
- $r = 0$: No linear relationship

Sensitive to outliers! Always visualize data before computing.

Spearman & Kendall Rank Correlation

Spearman's ρ:

- Pearson correlation applied to ranks of data.
- Measures *monotonic* relationships (not strictly linear).
- Formula: $1 - \frac{6\sum d_i^2}{n(n^2-1)}$ (where d_i is rank difference).

Kendall's τ:

- Based on concordant and discordant pairs.
- More robust for small sample sizes with many ties.

Interpretation Guidelines

Magnitude of $|r|$:

- $0.00 - 0.19$: Very weak
- $0.20 - 0.39$: Weak
- $0.40 - 0.59$: Moderate
- $0.60 - 0.79$: Strong
- $0.80 - 1.00$: Very strong

Context Matters:

- In physics, $r = 0.9$ might be poor.
- In social sciences, $r = 0.5$ might be excellent.

Data Overview: `mtcars` Dataset

We will use the built-in `mtcars` dataset (1974 Motor Trend US Magazine).

```r
# Load data
data(mtcars)

# View structure
str(mtcars)
# 'data.frame': 32 obs. of 11 variables:
#  $ mpg : num  21 21 22.8 21.4 18.7...
#  $ cyl : num  6 6 4 6 8...
#  $ disp: num  160 160 108 258 360...
#  $ hp  : num  110 110 93 110 175...
#  $ wt  : num  2.62 2.87 2.32 3.21 3.44...
```

Variables of Interest:

- `mpg`: Miles per gallon (Outcome)
- `wt`: Weight (Predictor)
- `hp`: Horsepower (Predictor)

Visualizing Relationships

Scatterplots are essential before correlation.

```r
library(ggplot2)

# Scatterplot: Weight vs MPG
ggplot(mtcars, aes(x=wt, y=mpg)) +
  geom_point(color="blue", size=3) +
  geom_smooth(method="lm", se=FALSE, color="red") +
  labs(title="Relationship between Weight and MPG",
       x="Weight (1000 lbs)", y="Miles per Gallon")
```

Observation: Appears to be a strong negative linear relationship.

Computing Pearson Correlation in R

```r
# Compute Pearson correlation
cor(mtcars$wt, mtcars$mpg, method = "pearson")
# [1] -0.8676594

# Test significance
cor.test(mtcars$wt, mtcars$mpg, method = "pearson")
# t = -9.559, df = 30, p-value = 1.294e-10
# 95% CI: [-0.9338, -0.7441]
```

Interpretation:

- $r = -0.87$: Strong negative correlation.
- $p < 0.001$: Statistically significant.
- Heavier cars tend to have lower fuel efficiency.

Computing Spearman Correlation in R

Use when normality is violated or data is ordinal.

```r
# Compute Spearman correlation
cor(mtcars$wt, mtcars$mpg, method = "spearman")
# [1] -0.8869813

# Test significance
cor.test(mtcars$wt, mtcars$mpg, method = "spearman")
# S = 3542, p-value = 1.488e-11
# rho = -0.887
```

Note: Spearman ρ is slightly higher here, suggesting a robust monotonic relationship.

Correlation Matrix

Examine relationships between multiple variables simultaneously.

```r
# Select numeric columns
num_vars <- mtcars[, c("mpg", "cyl", "disp", "hp", "wt")]

# Correlation matrix
cor_matrix <- cor(num_vars, method = "pearson")
print(round(cor_matrix, 2))

#            mpg   cyl  disp   hp   wt
# mpg     1.00 -0.85 -0.85 -0.78 -0.87
# cyl    -0.85  1.00  0.90  0.83  0.78
```

Insight: `cyl`, `disp`, `hp`, and `wt` are all highly correlated with `mpg`.

Visualization: Corrplot

Heatmaps provide intuitive visual summary.

```r
library(corrplot)

# Visualize correlation matrix
corrplot(cor_matrix, method = "color",
         type = "upper", tl.cex = 0.8,
         title = "Correlation Matrix of Car Variables",
         mar = c(0,0,1,0))
```

- Darker blue = Strong positive
- Darker red = Strong negative
- Size of circle = Magnitude

Introduction to Regression

Definition: A statistical method to model the relationship between a dependent variable (Y) and one or more independent variables (X).

Goals:

1. **Prediction:** Estimate Y for new X values.
2. **Inference:** Understand the strength and nature of relationships.

Types:

- Simple Linear Regression (1 predictor)
- Multiple Linear Regression (≥ 2 predictors)

Simple Linear Regression Model

Equation:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Y_i: Dependent variable (Response)
- X_i: Independent variable (Predictor)
- β_0: Intercept (Value of Y when $X = 0$)
- β_1: Slope (Change in Y per 1-unit change in X)
- ε_i: Error term (Residual) $\sim N(0, \sigma^2)$

Assumption: The relationship is linear in parameters.

Least Squares Method

Objective: Find $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the Sum of Squared Errors (SSE).

$$SSE = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n}(Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2$$

Solutions:

- $\hat{\beta}_1 = r\frac{S_y}{S_x}$
- $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X}$

R uses numerical optimization to solve this efficiently.

Coefficients Interpretation

Intercept (β_0):

- Expected value of Y when $X = 0$.
- Caution: Only meaningful if $X = 0$ is within the range of data.

Slope (β_1):

- Expected change in Y for a 1-unit increase in X, holding others constant.
- Sign indicates direction (+ or -).
- Magnitude indicates effect size.

R-Squared Explained

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

- SST: Total Sum of Squares (Total variability)
- SSR: Regression Sum of Squares (Explained variability)
- SSE: Error Sum of Squares (Unexplained variability)

Interpretation:

- Proportion of variance in Y explained by X.
- Range: 0 to 1 (e.g., 0.75 means 75% explained).

Regression Assumptions (LINE)

For valid inference, four key assumptions must hold:

1. **L**inearity: Relationship between X and Y is linear.
2. **I**ndependence: Observations are independent (no autocorrelation).
3. **N**ormality: Residuals are normally distributed.
4. **E**qual Variance (Homoscedasticity): Residual variance is constant across X.

Violations lead to biased coefficients or invalid p-values.

Assumption: Linearity

Check: Scatterplot of X vs Y or Residuals vs Fitted.

- **Valid:** Points scattered randomly around zero line.
- **Violation:** U-shaped or curved pattern.

Remedy:

- Transform variables (log, sqrt).
- Add polynomial terms (X^2).
- Use non-linear models.

Assumption: Independence

Check: Study design (random sampling) or Durbin-Watson test (for time series).

- **Valid:** No pattern in residual order.
- **Violation:** Autocorrelation (common in time-series data).

Remedy:

- Use Time Series models (ARIMA).
- Mixed-effects models for clustered data.

Assumption: Normality

Check: Q-Q Plot of residuals or Shapiro-Wilk test.

- **Valid:** Points follow the diagonal line.
- **Violation:** Points deviate at tails (S-shape).

Remedy:

- Transform response variable (Y).
- Remove outliers.
- Use robust regression.

Assumption: Homoscedasticity

Check: Scale-Location Plot or Breusch-Pagan Test.

- **Valid:** Random scatter with constant spread.
- **Violation:** Funnel shape (variance increases with fitted values).

Remedy:

- Transform Y (log transformation often stabilizes variance).
- Weighted Least Squares (WLS).

SLR in R: `lm()` Function

```r
# Fit Simple Linear Regression
# Model: mpg predicted by weight (wt)
model_slr <- lm(mpg ~ wt, data = mtcars)

# View model object
model_slr
# Call: lm(formula = mpg ~ wt, data = mtcars)
# Coefficients:
# (Intercept)           wt
#      37.285       -5.344
```

Equation: $\widehat{mpg} = 37.29 - 5.34 \times wt$

Model Summary

```r
# Detailed statistics
summary(model_slr)

# Key outputs:
# Coefficients:
#             Estimate Std. Error t value Pr(>|t|)
# (Intercept)  37.2851     1.8776   19.86   <2e-16 ***
# wt           -5.3445     0.5591   -9.56   1.29e-10 ***
# ---
# Residual standard error: 3.046 on 30 degrees of freedom
# Multiple R-squared:  0.7528, Adjusted R-squared:  0.7446
# F-statistic: 91.38 on 1 and 30 DF, p-value: 1.294e-10
```

Confidence Intervals

Uncertainty around coefficients.

```r
# 95% Confidence Intervals for coefficients
confint(model_slr, level = 0.95)
#                  2.5 %    97.5 %
# (Intercept) 33.450500 41.119681
# wt          -6.489225 -4.199756
```

Interpretation:

- We are 95% confident the true slope lies between -6.49 and -4.20.
- Since 0 is not in the interval, the effect is significant.

Prediction

Predict MPG for a car weighing 3.5 tons.

```r
# New data frame
new_car <- data.frame(wt = 3.5)

# Predict with Confidence Interval
predict(model_slr, newdata = new_car,
        interval = "confidence")
#       fit      lwr      upr
# 1 18.57939 17.39436 19.76442

# Predict with Prediction Interval (wider)
predict(model_slr, newdata = new_car,
        interval = "prediction")
#       fit      lwr      upr
# 1 18.57939 12.29293 24.86585
```

Multiple Linear Regression

Model: Include multiple predictors (e.g., weight + horsepower).

```r
# Fit Multiple Linear Regression
model_mlr <- lm(mpg ~ wt + hp, data = mtcars)

# Summary
summary(model_mlr)
# Coefficients:
#             Estimate Std. Error t value Pr(>|t|)
# (Intercept) 37.22727    1.59879  23.285  < 2e-16 ***
# wt          -3.87783    0.63273  -6.129 1.12e-06 ***
# hp          -0.03177    0.00903  -3.519  0.00145 **
# Multiple R-squared:  0.8268, Adjusted R-squared:  0.815
```

Interpretation: Multiple Coefficients

Equation: $\widehat{mpg} = 37.23 - 3.88(wt) - 0.032(hp)$

Interpretation of `wt`:

- For every 1-unit increase in weight, MPG decreases by 3.88, holding horsepower constant.

Interpretation of `hp`:

- For every 1-unit increase in horsepower, MPG decreases by 0.032, holding weight constant.

Key: "Holding other variables constant" distinguishes MLR from SLR.

Adjusted R-Squared

Problem: R^2 always increases when adding predictors, even if useless.
Solution: Adjusted R^2 penalizes for adding unnecessary predictors.

$$R^2_{adj} = 1 - (1 - R^2)\frac{n-1}{n-p-1}$$

- n: Sample size
- p: Number of predictors

Use Adjusted R^2 when comparing models with different numbers of predictors.

Multicollinearity

Definition: High correlation between predictor variables.
Consequences:

- Unstable coefficient estimates.
- Inflated standard errors.
- Difficult to isolate individual effects.

Detection:

- Correlation matrix ($|r| > 0.8$).
- Variance Inflation Factor (VIF).

VIF Calculation

Rule of Thumb: VIF > 5 or 10 indicates problematic multicollinearity.

```r
library(car)

# Calculate VIF
vif(model_mlr)
#        wt        hp
# 3.345678 3.345678

# Interpretation: VIF < 5, so multicollinearity is acceptable.
```

Remedy if High:

- Remove one of the correlated variables.
- Combine variables (PCA).
- Use Ridge/Lasso Regression.

Diagnostic Plots Overview

R provides 4 standard diagnostic plots.

```r
# Generate diagnostic plots
par(mfrow=c(2,2))
plot(model_mlr)
par(mfrow=c(1,1))
```

1. Residuals vs Fitted
2. Normal Q-Q
3. Scale-Location
4. Residuals vs Leverage

Residuals vs Fitted

Purpose: Check Linearity and Homoscedasticity.

- **Ideal:** Random scatter around horizontal line at 0.
- **Curve:** Indicates non-linearity.
- **Funnel:** Indicates heteroscedasticity.

In `mtcars` model: Points are fairly random, suggesting linearity holds.

Normal Q-Q Plot

Purpose: Check Normality of Residuals.

- **Ideal:** Points fall along the dashed diagonal line.
- **Deviation:** S-shape indicates skewness; curves indicate kurtosis.

In `mtcars` model: Points follow line closely, normality assumption met.

Scale-Location Plot

Purpose: Check Homoscedasticity (Equal Variance).

- Y-axis: $\sqrt{|StandardizedResiduals|}$
- **Ideal:** Horizontal line with randomly spread points.
- **Violation:** Increasing trend (funnel).

In `mtcars` model: Slight spread but generally acceptable.

Cook's Distance

Purpose: Identify Influential Observations.

- Measures how much coefficients change if observation is removed.
- **Threshold:** $D_i > 4/n$ or $D_i > 1$.

Action:

- Investigate outliers (data entry error?).
- Run model with and without outlier to check stability.

Formal Tests (Shapiro, BP)

Supplement plots with statistical tests.

```r
# 1. Normality: Shapiro-Wilk
shapiro.test(residuals(model_mlr))
# p > 0.05 -> Normality assumed

# 2. Homoscedasticity: Breusch-Pagan
library(lmtest)
bptest(model_mlr)
# p > 0.05 -> Homoscedasticity assumed

# 3. Independence: Durbin-Watson
dwtest(model_mlr)
# p > 0.05 -> No autocorrelation
```

Remedial Measures

If Assumptions Fail:

1. **Non-Linearity:** Add polynomial terms (‘poly(x, 2)‘) or transform variables.
2. **Heteroscedasticity:** Log-transform response (‘log(y)‘).
3. **Non-Normality:** Transform response or use robust regression (‘rlm‘).
4. **Outliers:** Verify data, consider removal if justified.
5. **Multicollinearity:** Remove variables or use regularization.

Reporting Results (APA)

Example Statement:

"A multiple linear regression was calculated to predict MPG based on car weight and horsepower. A significant regression equation was found ($F(2, 29) = 69.28, p < .001$), with an R^2 of .827. Car weight ($\beta = -3.88, p < .001$) and horsepower ($\beta = -0.03, p = .001$) were significant predictors."

Always Include:

- F-statistic, degrees of freedom, p-value.
- R^2 and Adjusted R^2.
- Coefficients, Standard Errors, and Confidence Intervals.

Common Pitfalls

- **Causation:** Assuming regression implies causality (it does not).
- **Extrapolation:** Predicting outside the range of observed data.
- **Overfitting:** Too many predictors for sample size ($n < 10p$).
- **Ignoring Assumptions:** Reporting p-values without checking LINE.
- **Data Leakage:** Using future data to predict past events.

Statistical rigor protects against misleading conclusions.

Summary & Resources

Key Takeaways:

- Correlation measures association; Regression models prediction.
- Always visualize data before analysis.
- Verify LINE assumptions using plots and tests.
- Report effect sizes and confidence intervals, not just p-values.

Further Learning:

- James, G. et al. (2021). *An Introduction to Statistical Learning*.
- CDAM Resources: `https://cdam.chuka.ac.ke`

Thank you for your attention!